

GNU nano 8.7.1

task1.c

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

int main() {
    char command[100];

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid_t pid = fork();

    if (pid < 0) {
        printf("Fork failed!\n");
        return 1;
    }

    else if (pid == 0) {
        printf("Child Process\n");
        printf("Child PID : %d\n", getpid());

        printf("Executing command: %s\n", command);

        execvp(command, command, NULL);

        printf("Command execution failed!\n");
    }

    else {
        printf("Parent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        wait(NULL);

        printf("Child process completed.\n");
    }
}
```

```
shreyaj@LAPTOP-FT6VVM4D: ~$ uname -a
Linux LAPTOP-FT6VVM4D 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
shreyaj@LAPTOP-FT6VVM4D: ~$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:               39 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      12
On-line CPU(s) list:        0-11
Vendor ID:                   GenuineIntel
Model name:                  13th Gen Intel(R) Core(TM) i5-13420H
CPU family:                  6
Model:                      186
Thread(s) per core:         2
Core(s) per socket:         6
Socket(s):                   1
Stepping:                    2
BogoMIPS:                    5222.42
Flags:                       fpu vme de pse tsc msr pae mce cx8 apic sep mtr
                             r pge mca cmov pat pse36 clflush mmx fxsr sse s
                             se2 ss ht syscall nx pdpe1gb rdtscp lm constant
                             _tsc rep_good nopl xtopology tsc_reliable nonst
                             op_tsc cpuid tsc_known_freq pni pclmulqdq vmx s
                             sse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe p
                             opcnt tsc_deadline_timer aes xsave avx f16c rdr
                             and hypervisor lahf_lm abm 3dnowprefetch ssbd i
                             brs ibpb stibp ibrs_enhanced tpr_shadow ept vpi
                             d ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi
                             2 erms invpcid rdseed adx smap clflushopt clwb
                             sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni
                             vnmi umip waitpkg gfni vaes vpclmulqdq rdpid mo
                             vdiri movdir64b fsrm md_clear serialize ibt flu
                             sh_l1d arch_capabilities

Virtualization features:
Virtualization:              VT-x
Hypervisor vendor:           Microsoft
Virtualization type:         full
Caches (sum of all):
L1d:                         288 KiB (6 instances)
L1i:                         192 KiB (6 instances)
L2:                           7.5 MiB (6 instances)
L3:                          12 MiB (1 instance)
NUMA:
NUMA node(s):                1
NUMA node0 CPU(s):          0-11
```

```
shreyaj@LAPTOP-FT6VVM  x  Ubuntu  x  Ubuntu  x  Ubuntu  x  Ubuntu  x  Ubuntu  x  +  -  X
Vulnerabilities:
Gather data sampling:    Not affected
Ghostwrite:             Not affected
Indirect target selection: Not affected
Itlb multihit:          Not affected
L1tf:                   Not affected
Mds:                    Not affected
Meltdown:               Not affected
Mmio stale data:        Not affected
Old microcode:          Not affected
Reg file data sampling:  Mitigation; Clear Register File
Retbleed:               Mitigation; Enhanced IBRS
Spec rstack overflow:    Not affected
Spec store bypass:       Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1:             Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2:             Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_S
Srbds:                  Not affected
Tsa:                    Not affected
Tsx async abort:        Not affected
Vmscape:                Not affected
shreyaj@LAPTOP-FT6VVM4D:~$ ps
  PID TTY          TIME CMD
 14440 pts/6    00:00:00 bash
 14491 pts/6    00:00:00 ps
```

```
shreyaj@LAPTOP-FT6VVM4D:~$ top
top - 15:07:15 up 1:19, 1 user, load average: 0.00, 0.02, 0.07
Tasks: 43 total, 1 running, 42 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.2 us, 0.3 sy, 0.0 ni, 99.5 id, 0.0 wa, 0.0 hi, 0.1 si, 0.0 st
MiB Mem : 7776.6 total, 7061.2 free, 653.0 used, 229.5 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7123.7 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
113	root	20	0	35376	12332	9232	S	0.3	0.2	0:03.33	systemd-udevd
14503	shreyaj	20	0	8320	6148	3952	R	0.3	0.1	0:00.06	top
1	root	20	0	24440	15356	11132	S	0.0	0.2	0:03.63	systemd
2	root	20	0	3180	2212	2072	S	0.0	0.0	0:00.02	init-systemd(Ub
7	root	20	0	3228	2160	2008	S	0.0	0.0	0:00.63	init
72	root	19	-1	50400	17052	15704	S	0.0	0.2	0:00.47	systemd-journal
107	systemd+	20	0	22540	14580	11956	S	0.0	0.2	0:00.15	systemd-resolve
135	root	20	0	2888	1948	1820	S	0.0	0.0	0:00.12	chronyd-starter
136	root	20	0	4472	3172	2924	S	0.0	0.0	0:00.02	cron
137	message+	20	0	9040	5856	4780	S	0.0	0.1	0:00.73	dbus-daemon
143	root	20	0	44092	29724	14576	S	0.0	0.4	0:00.85	networkd-dispat
175	root	20	0	18644	9512	8184	S	0.0	0.1	0:00.49	systemd-logind
265	_chrony	20	0	20420	6356	5584	S	0.0	0.1	0:00.17	chronyd
277	_chrony	20	0	12092	2464	1824	S	0.0	0.0	0:00.01	chronyd
311	root	20	0	123456	32700	18092	S	0.0	0.4	0:00.52	unattended-upgr
313	root	20	0	5492	2740	2556	S	0.0	0.0	0:00.01	agetty
366	root	20	0	8648	5576	4732	S	0.0	0.1	0:00.01	login
408	shreyaj	20	0	22468	12744	10300	S	0.0	0.2	0:00.37	systemd
412	shreyaj	20	0	22908	4088	2088	S	0.0	0.1	0:00.00	(sd-pam)
436	shreyaj	20	0	6136	5420	3864	S	0.0	0.1	0:00.03	bash
772	root	20	0	3184	1124	980	S	0.0	0.0	0:00.00	SessionLeader
773	root	20	0	3200	1260	1108	S	0.0	0.0	0:00.09	Relay(778)
778	shreyaj	20	0	6268	5544	3872	S	0.0	0.1	0:00.17	bash
1258	root	20	0	3184	1124	980	S	0.0	0.0	0:00.00	SessionLeader
1259	root	20	0	3200	1260	1108	S	0.0	0.0	0:00.11	Relay(1264)
1264	shreyaj	20	0	6268	5608	3936	S	0.0	0.1	0:00.25	bash
1752	root	20	0	3184	1124	980	S	0.0	0.0	0:00.00	SessionLeader
1753	root	20	0	3200	1260	1108	S	0.0	0.0	0:00.02	Relay(1758)
1758	shreyaj	20	0	6268	5608	3936	S	0.0	0.1	0:00.11	bash
1941	polkitd	20	0	306444	8560	7696	S	0.0	0.1	0:00.19	polkitd
2075	root	20	0	3184	1124	980	S	0.0	0.0	0:00.00	SessionLeader
2076	root	20	0	3200	1260	1108	S	0.0	0.0	0:00.03	Relay(2083)
2083	shreyaj	20	0	6268	5544	3872	S	0.0	0.1	0:00.13	bash
3234	syslog	20	0	220548	5464	4616	S	0.0	0.1	0:00.06	rsyslogd
4971	shreyaj	20	0	8544	4380	3956	S	0.0	0.1	0:00.01	dbus-daemon
14105	root	20	0	3184	1124	980	S	0.0	0.0	0:00.00	SessionLeader
14106	root	20	0	3200	1260	1108	S	0.0	0.0	0:00.04	Relay(14110)

```
shreyaj@LAPTOP-FT6VVM: ~$ mkdir OS_LAB
mkdir: OS_LAB: File exists
shreyaj@LAPTOP-FT6VVM: ~$ cd OS_LAB
shreyaj@LAPTOP-FT6VVM: ~/OS_LAB$ nano task1.c
shreyaj@LAPTOP-FT6VVM: ~/OS_LAB$ ^C
shreyaj@LAPTOP-FT6VVM: ~/OS_LAB$ gcc task1.c -o task1
shreyaj@LAPTOP-FT6VVM: ~/OS_LAB$ sudo apt update
sudo apt install build-essential
[sudo: authenticate] Password:
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
25 packages can be upgraded. Run 'apt list --upgradable' to see them.
build-essential is already the newest version (12.12ubuntu2.26.04.2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 25
shreyaj@LAPTOP-FT6VVM: ~/OS_LAB$ ./task1
Enter a Linux command: ls
Parent Process
Parent PID : 14386
Child PID  : 14387
Child Process
Child PID : 14387
Executing command: ls
task1 task1.c
Child process completed.
shreyaj@LAPTOP-FT6VVM: ~/OS_LAB$ pwd
/home/shreyaj/OS_LAB
shreyaj@LAPTOP-FT6VVM: ~/OS_LAB$
```